ChatGPT

From Connectome Spectra to Clinical State Space in Depression

Executive summary

A unifying way to relate *graph-theoretic* descriptions of brain connectivity to *dynamical-systems* "state space" descriptions is to treat brain activity as a trajectory on a structured manifold whose **coordinates are modal weights**—either weights on **graph/Laplacian eigenmodes** (connectome eigenmodes, connectome harmonics; sometimes also geometric eigenmodes) or weights on **spatiotemporal wave bases** (cortical traveling waves, phase-gradient modes). In this view, changes in static/dynamic functional connectivity (FC), traveling-wave statistics, and spectral content correspond to changes in (i) how trajectories explore modal-weight space and (ii) the stability geometry of that space (basins, barriers, and "landscape curvature"). [1]

Several primary-source frameworks formalize parts of this mapping. **Connectome-harmonic / graph-Fourier approaches** define an anatomically grounded eigenbasis (graph Laplacian eigenvectors) and explicitly interpret time-varying mode activations as a repertoire of "building blocks" for whole-brain patterns. [2] **Analytic spectral structure–function links** show that a small number of structural Laplacian eigenmodes can reconstruct FC, implying that shifts in FC can be represented as shifts in eigenmode contributions. [3] **Neural-field / wave accounts** argue that large-scale activity is well captured by eigenmodes of the cortical sheet (Laplace–Beltrami/geometric eigenmodes) and their wave dynamics, creating a direct theoretical bridge between eigenmodes and cortical traveling waves. [4] **Energy-landscape methods** (maximum entropy / Ising-style) estimate attractor-like minima from data, providing operational quantities like basin depth and transition structure. [5]

In depression, large-scale connectivity abnormalities (especially involving default-mode and control networks) are robust at the group level but not always directionally uniform across subtypes, chronicity, and medication. [6] Dynamic FC studies increasingly support the idea that depression involves altered dwell times/occupancies of recurrent FC configurations—i.e., altered exploration of state space. [7]

Treatments can be interpreted as **perturbations that reshape the effective landscape**. In serotonergic psychedelic therapy, multiple studies report decreased modularity and increased global integration (a "loosening" of network segregation), consistent with theoretical proposals of reduced landscape curvature / flattened minima and increased exploratory capacity. [8] In rTMS, both large-cohort rs-fcMRI analyses and mechanistic whole-brain modeling indicate treatment-linked changes in variability/metastability and separable dynamical subtypes, which naturally map onto different attractor regimes. [9] In ECT, modular reorganization and increased inter-module coupling in emotion-related networks are frequently reported, suggesting a large perturbation that can move the system between basins—though explicit attractor-landscape quantification remains less standardized. [10]

Conceptual and mathematical foundations

Static and dynamic functional connectivity

Static FC (sFC) is typically an undirected association matrix $F \in \mathbb{R}^{N\times N}$ computed across an interval $t = 1, \ldots, T$, most often via Pearson correlation (or related dependence measures) between regional time series $x_i(t)$ and $x_j(t)$:

$$F_{ij} = \mathrm{corr}(x_i, x_j) \quad \text{(or Fisher-}z\text{ transformed)}.$$

This "whole-window" abstraction is ubiquitous in depression research and meta-analyses. [11]

Dynamic FC (dFC) replaces the single matrix F by a time-indexed family $F(t)$ (or windowed $F^{(w)}$), estimated via sliding windows, time–frequency coherence, phase-coherence connectivity, hidden Markov models, or other approaches. [12] A key technical point for interpretation is that some dFC estimators can confound neural dynamics with sampling variability, motion artifacts, and the temporal smoothing imposed by fMRI preprocessing and hemodynamics. [13]

A particularly relevant eigenvector-based dFC estimator is **LEiDA**, which uses the *leading eigenvector* of the instantaneous BOLD phase-coherence matrix (single-TR resolution) to characterize recurrent phase-locking patterns/states. [14]

Cortical traveling waves

A **cortical traveling wave** is a spatiotemporal pattern where a phase (or activation) field $\phi(\mathbf{r}, t)$ propagates across cortex with a nonzero spatial phase gradient, often modeled locally as

$$\phi(\mathbf{r}, t) \approx \mathbf{k} \cdot \mathbf{r} - \omega t + \phi_0,$$

so that wave direction is along $\mathbf{k}$ and speed $v = \omega/\|\mathbf{k}\|$. Empirically, traveling-wave detection often uses Hilbert-transform phases and fits phase gradients over sensor/source space. [15]

Human intracranial recordings show that theta/alpha oscillations frequently form traveling waves, and modeling work suggests weakly coupled oscillator mechanisms and frequency gradients as drivers of propagation direction. [16] Connectome-based simulations can produce traveling waves whose phase patterns align with resting-state networks and can be interpreted as sequencing/hierarchical coordination mechanisms. [17]

Graph Laplacian, eigenmodes, and connectome harmonics

Given a (usually undirected, weighted) structural connectome adjacency matrix A, degree matrix D, and (one choice of) graph Laplacian L, a standard definition is

$$L = D - A,$$

with normalized variants also common. [18] The **eigendecomposition**

$$LU = U\Lambda$$

yields eigenvectors $U = [\psi_1, \ldots, \psi_N]$ (graph eigenmodes) and eigenvalues $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$. [19]

In **graph signal processing**, the Laplacian eigenvectors define a **graph Fourier transform**: any cortical/parcel signal $x(t) \in \mathbb{R}^N$ can be expanded as

$$x(t) = \sum_{k=1}^{N} a_k(t)\,\psi_k \quad \text{or} \quad a(t) = U^\top x(t).$$

[20]

In the connectome-harmonic literature, Laplacian eigenvectors of a high-resolution connectome are called **connectome harmonics** and are treated as an anatomically grounded extension of the Fourier basis for cortex-wide activity. [21]

Attractors, landscapes, and "rigidity"

A brain "state space" description starts from a dynamical system

$$\dot{x} = f(x; \theta) + \eta(t),$$

where x is a neural state (or latent state), θ parameters, and η noise. **Attractors** are invariant sets (fixed points, limit cycles, etc.) toward which nearby trajectories converge; a **basin of attraction** is the set of initial conditions that flow to that attractor. [22]

An **attractor landscape** is a metaphor made operational in several ways. One operational route is an **energy landscape** estimated from data (often fMRI) using pairwise maximum entropy / Ising-style models, where local minima correspond to attractor-like states and barrier structure relates to transition kinetics. [23] A complementary theoretical route (used explicitly in psychedelic theory) links "landscape curvature" to precision/confidence of high-level beliefs: reduced precision corresponds to reduced curvature/flattened minima, enabling easier basin escape. [24]

Mapping eigenmodes and traveling-wave bases into state-space coordinates

The modal-coordinate construction

The core mathematical proposal—shared across connectome-harmonic, spectral graph, neural-field, and wave-eigenmode frameworks—is:

1. Choose a structural (or geometric) operator L whose eigenvectors ψ_k form a basis on cortex (graph Laplacian) or cortical manifold (Laplace–Beltrami). [25]
2. Project observed activity $x(t)$ onto that basis: $a_k(t) = \langle x(t), \psi_k \rangle$. [26]
3. Treat the **weight vector** $a(t) = [a_1(t), \ldots, a_K(t)]$ (with $K \ll N$ for truncation) as a **low-dimensional coordinate chart** for brain state space. [27]

Why this gives a state-space link: if the underlying dynamics are a neural field or diffusion-/wave-like process on the connectome/cortex, then substitution of the eigen-expansion into the field equation yields coupled ODEs (or SDEs) for the coefficients $a_k(t)$. In certain linearized regimes, modes approximately decouple, turning the system into a set of low-dimensional modes with stability governed by eigenvalues and biophysical parameters—i.e., an explicit dynamical system in modal coordinates. [28]

How FC changes become landscape deformations in modal space

There are two complementary mappings from “connectivity changes” to “landscape changes”:

Mapping A: FC eigenstructure $\rightarrow$ state-space geometry.
If FC can be reconstructed from a small number of structural Laplacian eigenmodes (and eigenvalues) or if FC and SC share eigenvectors under a diffusion-like mapping, then treatment-induced changes in FC can be represented as shifts in the effective eigenvalue weighting of modes (“which eigenmodes matter more”), which changes the effective dimensionality and the stability structure of trajectories in modal space. [29]

Mapping B: probability of visiting patterns $\rightarrow$ quasi-potential / energy landscape.
If we model the coarse-grained dynamics as noise-driven motion among metastable regions, then state occupancy probabilities $p(s)$ (for discrete states s) imply an “energy” $E(s) \propto -\log p(s)$. Pairwise maximum entropy models fit parameters h_i, J_{ij} such that $p(s)$ matches empirical statistics; these parameters define an explicit energy landscape whose minima correspond to stable states and whose barriers constrain transitions. [23]

Interpretation of “deep/rigid attractors”: in either mapping, “rigidity” corresponds to (i) higher barrier heights / deeper minima (reduced transitions) and/or (ii) stronger effective restoring forces (larger negative real parts of Jacobian eigenvalues in the local linearization). Psychedelic theory explicitly frames therapeutic action as reducing curvature/flattening minima to permit escape from pathological basins (e.g., depressive rumination loops). [24]

Traveling waves as structured trajectories in eigenmode subspaces

Traveling waves are not merely “another feature”; they can be read as **specific trajectories** in modal space:

- In wave equations on cortex, eigenmodes correspond to standing-wave spatial patterns; traveling waves can be represented by superpositions of (often near-degenerate) eigenmodes with relative phase shifts, producing apparent propagation. [30]
- In connectome-informed oscillator networks, traveling waves emerge in intermediate coupling regimes (between incoherence and full synchrony) and co-occur with frequency gradients; these regimes can be interpreted as operating near a transition boundary in state space (a “working point” where metastability is large). [31]

Thus, changes in traveling-wave prevalence/direction/speed can be interpreted as changes in (i) coupling strengths and delays (control parameters) and (ii) which modal subspaces are activated—i.e., a concrete bridge between wave statistics and attractor-regime location.

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flowchart LR
  A[Structural connectome / cortical geometry] --> B[Operator L: graph Laplacian or Laplace–Beltrami]
  B --> C[Eigenmodes ψ_k and eigenvalues λ_k]
  C --> D[Project data: a_k(t)=<x(t),ψ_k>]
  D --> E[Trajectory a(t) in modal-weight state space]
  E --> F[Estimate landscape: basins, barriers, curvature]
  F --> G[Clinical phenotype: rigidity vs flexibility]
  H[Treatment perturbation: ECT / rTMS / psychedelics] --> I[Parameter change: coupling, excitation-inhibition, noise, precision]
  I --> F
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The conceptual mapping above is directly motivated by eigenmode-basis claims (connectome harmonics; graph neural fields; geometric eigenmodes) and landscape-flattening claims in psychedelic theory. [32]

Frameworks and mechanistic models instantiating the mapping

Comparative table of major frameworks

Framework / exemplar primary source	Key proponents	Core idea	Mathematical mapping to state space	Empirical/clinical support relevant to depression & treatment	Pros	Cons / limitatior
Connectome harmonics / connectome-specific harmonic waves (Nat Commun 2016)	Selen Atasoy [33]	Structural Laplacian eigenvectors form a universal basis; resting-state networks align with specific harmonics; neural-field dynamics can self-organize into these patterns. [34]	$x(t) = \sum_k a_k(t)\psi_k$; dynamics on cortex become dynamics of $\{a_k\}$ under neural-field models on the connectome. [34]	Provides the basis used in psychedelic connectome-harmonic decompositions; supports basis–network correspondence. [35]	Anatomically grounded coordinates; compressive representation; links to neural fields. [34]	Depends connecto estimatic choices; directed effective connectiv not captu by symm Laplaciar [34]
Structural Laplacian eigen-structure predicts FC (NeuroImage 2018)	Farrukh Abdelnour [36]; Ashish Raj [37]	Analytic relationship between SC Laplacian spectra and FC spectra; few eigenmodes reconstruct FC. [3]	FC is approximated by weighted sums of Laplacian eigencomponents; changes in FC ↔ changes in eigenvalue weighting. [3]	Supports spectral interpretations of altered FC in disease generally; provides mathematical bridge from connectivity to low-dimensional spectral coordinates. [3]	Closed form; fast; explicit eigen-relationship. [3]	Linearize diffusion assumpt limited fo strongly nonlinea directed dynamics

Framework / exemplar primary source	Key proponents	Core idea	Mathematical mapping to state space	Empirical/clinical support relevant to depression & treatment	Pros	Cons / limitation
Graph neural fields / CHAOSS (PLOS CB 2021)	Michel Aqil [38]	Implements neural-field-type stochastic integro-differential equations directly on connectome graphs, yielding analytic spectra, FC/coherence predictions. [39]	Graph Fourier/ Laplacian diagonalization yields mode-wise analytic predictions; equilibrium fluctuations and stability define effective attractor regimes in modal coordinates. [39]	Provides mechanistic route from connectome changes to FC and spectral signatures; useful template for simulating treatment-induced parameter shifts (E/I balance, coupling). [39]	Mechanistic and computationally efficient; explicit Laplacian machinery. [39]	Requires model-ch commitm (kernel fo observat model); paramete identifiab can be challengi [39]
Geometric eigenmodes and wave model (Nature 2023)	James C. Pang [40]	Cortical geometry eigenmodes (Laplace–Beltrami) parsimoniously reconstruct activity; wave dynamics explain spatiotemporal organization (including dFC benchmarks). [41]	$\Delta\psi = -\lambda\psi$ on cortical manifold; activity is weighted sum of geometric modes; wave PDE induces dynamics over coefficients. [41]	Suggests treatment effects could be framed as shifts in wave-model parameters vs network-coupled mass models; implies traveling-wave/eigenmode link. [41]	Strong generative grounding; links eigenmodes and waves; addresses FCD benchmarks. [41]	Connecto may still matter (subcorte long-rang specificit clinical translatio direct. [4

Framework / exemplar primary source	Key proponents	Core idea	Mathematical mapping to state space	Empirical/clinical support relevant to depression & treatment	Pros	Cons / limitation
Connectome topology directs traveling waves (Nat Commun 2024)	Dominik P. Koller [42]	Structural instrength gradients + delays can determine traveling-wave direction and frequency gradients; best fits MEG FC in intermediate regimes. [43]	Oscillator-network model with coupling scaling and conduction speed; traveling waves emerge between incoherence and synchrony—interpretable as regime boundaries in state space. [43]	Provides explicit parameter regimes where wave-like coordination appears; relevant to stimulation and disorders affecting coupling. [43]	Connects waves ↔ FC and connectome topology; highlights control parameters. [43]	Most evid in health modeling depressio specific evidence remains indirect.
Energy landscapes via maximum entropy (Front Neuroinform 2014; RSTA 2017)	Takamitsu Watanabe [44] ; Thilo Ezaki [45]	Fit maximum entropy models to network activity patterns; define energy landscape; minima ≈ attractor-like states. [46]	$E(s) = -\sum h_i s_i - \sum J_{ij} s_i s_j$; state transitions constrained by barriers; "rigidity" quantifiable via basin depth and transition graph. [47]	A direct operationalization of "attractor landscape" for neuroimaging; useful for testing treatment-induced flattening/ deepening hypotheses. [48]	Makes landscape concrete; yields testable barrier/dwell predictions. [49]	Binary/st discretiza choices n scaling to many reg is hard; interpret depends preproce [50]
REBUS / entropic brain (Pharm Rev 2019; Front Hum Neurosci 2014)	Robin L. Carhart-Harris [51] ; Karl J. Friston [52]	Psychedelics relax high-level prior precision, increasing entropy/ criticality and flattening landscapes, enabling escape from pathological basins (e.g., depression). [24]	Precision ↓ ↔ landscape curvature ↓ (flattened minima); increased repertoire and transitions ↔ shallower basins. [24]	Aligns with observed decreased modularity/ increased integration after psilocybin therapy and acute psychedelic FC changes. [53]	Clinically intuitive; connects computational theory to network measures. [24]	Abstract mapping "priors" t measura neural m is not un risks pos interpret without e generativ models.

Framework / exemplar primary source	Key proponents	Core idea	Mathematical mapping to state space	Empirical/clinical support relevant to depression & treatment	Pros	Cons / limitation
Whole-brain model + metastable substates + perturbation (Sci Rep 2017; HBM 2022; Brain Comms 2024)	Joana Cabral [54]; Joseph Vohryzek [55]	Use leading-eigenvector phase-locking states to define metastable substates; fit whole-brain models; perturb models to drive transitions and infer causal targets. [56]	Empirical PMS/ LEiDA occupancy = coarse state-space distribution; model parameters (global coupling, Hopf bifurcation) control regime and transition ability. [57]	Directly applied to depression and psilocybin response prediction; identifies perturbation targets and links to receptor maps. [58]	Strong bridge from data → state space → intervention; yields testable stimulation predictions. [58]	Coarse-graining clusters a choice of matters; temporal limits; structura connecto often template based. 5
rTMS: dynamical subtypes & attractor regimes (PLOS CB 2023; JPN 2024)	Nicolas Kaboodvand [59]; Vinh Tan [60]	Whole-brain oscillatory models reveal distinct depression attractor regimes predicting rTMS response; large samples show rTMS alters rsFC variability in key regions. [61]	Model parameters define stable-focus/oscillatory regimes and metastability; empirical rsFC variability can be read as exploration breadth in state space. [61]	Depression subtyping by attractor dynamics; rTMS linked to changes in sgACC/ precuneus variability. [61]	Mechanistic stratification; explicit attractor portraits; clinically actionable. [62]	Generaliz across protocols targets n replicatio model identifiab and multimod limits. 6

Plain-language summaries of the main approaches

Eigenmode / harmonic coordinate approaches (connectome harmonics, geometric eigenmodes): the brain's anatomy defines a set of natural "vibration patterns." Any observed activity pattern is a mixture of these patterns. If depression or treatment changes which patterns dominate (or how fast and flexibly their weights change), that is a mathematically precise way to say "the brain moved in state space." [63]

Traveling-wave approaches: instead of focusing on correlations, quantify how phase patterns move across cortex. Changes in wave directionality/speed/coherence imply changes in coupling, delays, and hierarchy—i.e., changes in the system's operating regime. [64]

Energy-landscape approaches (maximum entropy): estimate which whole-brain activation patterns are stable and how hard it is to transition between them; depression can be tested as having deeper basins / higher barriers, while treatments can be tested as flattening or opening new low-energy paths. [23]

Whole-brain generative models + perturbation: fit a mechanistic model to an individual's (or group's) dynamics, then perturb parameters or nodes in silico to see what would be required to move from a depressive-state distribution to a healthier one—turning "landscape deformation" into explicit parameter changes and predicted stimulation targets. [65]

Empirical evidence in depression and interventions

Depression: connectivity and dynamical signatures consistent with altered "exploration" of state space

A large resting-state FC meta-analysis supports the broad claim that major depression involves dysconnectivity among large-scale networks (default-mode, frontoparietal control, affective networks), though the direction and localization of effects vary across studies. [66] Large multi-site work also highlights that DMN findings can differ by recurrence status and medication exposure, emphasizing heterogeneity and the risk of assuming a single "depression FC signature." [67]

Dynamic analyses strengthen the state-space framing: dFC studies and consortium-scale analyses report altered time-varying network configurations in MDD, consistent with the idea that depression is associated with altered dwell times/occupancy of recurrent connectivity patterns. [68] Mechanistic work that explicitly defines probabilistic metastable substates (via leading-eigenvector phase locking) observes group differences in substate probabilities and then uses whole-brain models to reproduce and perturb these distributions—directly operationalizing "moving between depressive and healthy state spaces." [69]

Psychedelics: integration, repertoire expansion, and landscape-flattening interpretations

In depression trials of **psilocybin therapy**, fMRI network analysis shows that antidepressant response correlates with **decreased modularity** and increased global integration, implying reduced network segregation. [70] Early treatment-resistant depression fMRI work also reports post-treatment changes in resting-state measures, supporting the use of FC as a mechanistic readout. [71]

In healthy-subject psychedelic neuroimaging, **LSD** increases **global functional connectivity** and compromises modular/rich-club organization, with effects linked to subjective phenomena like ego dissolution. [72] The **connectome-harmonic decomposition** of LSD fMRI reports increased total energy, increased high-frequency harmonic activation, and an expanded repertoire of active harmonic brain states —directly aligning with a "broadened state space" interpretation in modal coordinates. [73]

Dynamic eigenvector-based analyses of **psilocybin infusion** show destabilization of specific recurrent phase-locking states (including a frontoparietal control-like pattern) and increased occurrence of globally coherent states, i.e., altered transition structure across a repertoire of metastable states. [74] Recent depression-focused whole-brain modeling work uses metastable substates plus **dynamic sensitivity analysis** to identify perturbations/regions enabling a transition from a depressive to a "healthier" substate distribution, with implicated regions correlating with serotonin receptor density maps—linking pharmacology, state dynamics, and causal targets. [75]

These empirical observations are commonly interpreted through theoretical proposals that psychedelics flatten or relax the constraints of high-level dynamics (reduced prior precision), allowing escape from rigid pathological basins and enabling cognitive/affective updating. [24] A key open issue is that many "flattening" claims remain metaphorical unless paired with explicit landscape estimation (e.g., maximum entropy energy landscapes) or explicit model-based potentials defined in modal coordinates. [76]

rTMS: connectivity changes, variability, and attractor-regime stratification

Large-sample analyses indicate that rTMS can change resting-state measures in clinically relevant networks and can increase **individual variability** (e.g., mean correlational distance) in regions such as sgACC and precuneus, suggesting altered dynamical breadth rather than a simple "normalization toward the mean." [77]

Mechanistic whole-brain modeling goes further: a randomized trial combining resting-state modeling with stimulation outcomes stratifies depressed patients into distinct latent subtypes characterized by different **attractor dynamics** and differences in metastability/synchrony at baseline, with subtype differences predicting differential response to accelerated theta-burst stimulation (dmPFC). [62] This is one of the most direct instantiations of the requested mapping: fitted model parameters define a dynamical regime (attractor portrait), and stimulation response relates to regime membership. [62]

Independent of depression, TMS-EEG work plus digital-twin Kuramoto modeling shows that stimulation can transiently reduce metastability (and increase coherence), consistent with the idea that a perturbation can temporarily collapse dynamics into a more ordered coordination pattern—i.e., a transient change in the effective attractor structure. [78]

ECT: modular reorganization and dynamic-connectome implications

Systematic review and meta-analytic work supports that ECT produces measurable resting-state connectivity changes in depression, though findings vary with methodology and sample. [79]

A concrete example using modularity analysis of an emotion regulation network reports that ECT increased intra- and inter-module functional interactions and that specific reconfigurations correlated with clinical and cognitive measures, supporting an interpretation of ECT as inducing network-level reorganization (a large perturbation that could move the system between basins). [80] More recent work focuses explicitly on **connectome dynamics** after ECT (dynamic connectome measures and links to outcome), indicating a trend toward dynamical-state interpretations even when attractor language is not formalized. [81]

A limitation for the specific eigenmode→attractor mapping is that ECT studies less often report results in modal coordinates (harmonic power spectra, eigenmode weights, traveling-wave statistics) or in explicit landscape parameters (barrier heights, basin depth), leaving a methodological gap that is experimentally addressable. [82]

Open questions, testable predictions, and recommended analyses

Central open questions

A fundamental identifiability question is whether changes in FC (especially dFC) reflect (i) genuine changes in underlying neural dynamics, (ii) changes in observation/noise processes, or (iii) both; dynamic FC reviews emphasize methodological and interpretive pitfalls that must be addressed for rigorous landscape claims. [13]

A second question is which structural basis is most appropriate for clinical state spaces: **connectome eigenmodes** (structural connectome), **geometric eigenmodes** (cortical sheet), or hybrid operators integrating geometry + long-range tracts; recent work shows geometry-based eigenmodes can outperform connectome-derived bases for reconstructing activity, implying that "coordinates of state space" may depend strongly on which operator defines the basis. [83]

A third question is how to validate "deep/rigid attractors" in depression without circularity: we need operational definitions (e.g., basin depth via energy landscapes; barrier heights via transition statistics; stability via model Jacobians) that make falsifiable predictions. [84]

Testable predictions that directly link graph measures to landscapes

Prediction 1 (modal repertoire restriction in depression): relative to controls, depressed individuals will show (in eigenmode/harmonic coordinates) a more concentrated distribution of power/energy over low-order modes and reduced temporal variability of mode weights $a_k(t)$, corresponding to fewer visited regions of modal-weight space. [85]

Prediction 2 (treatment as flattening / barrier reduction): effective treatments (psilocybin therapy, successful rTMS, successful ECT) will reduce estimated landscape barrier heights and/or increase transition rates among metastable states—measurable either via maximum-entropy energy landscapes or via increased switching/entropy in metastable substate sequences. [86]

Prediction 3 (wave regime shift as coupling-parameter shift): if traveling waves depend on intermediate coupling regimes, then treatments that alter effective coupling (e.g., through excitation–inhibition balance or plasticity) should shift wave prevalence/directionality measures in ways predictable by oscillator-network parameter sweeps; this is especially testable in MEG/EEG. [87]

Suggested experiments and analyses

Data types (minimum viable; extendable): - rs-fMRI (for FC, dFC, LEiDA, energy landscapes), ideally multi-session to estimate reliability. [88]
- Diffusion MRI (for SC Laplacian eigenmodes / connectome harmonics; individual-specific when possible). [89]
- MEG/EEG (for traveling waves, phase dynamics, spectral measures, and stimulation-locked effects). [90]
- If available: PET receptor maps (5-HT2A/5-HT1A) for psychedelic mechanisms and for constraining parameter priors in models. [91]

Preprocessing / representation choices that matter for the mapping: - Use robust fMRI preprocessing and motion control; dynamic FC literature stresses that artifacts can mimic state changes. [92]
- For eigenmode/harmonic analyses, explicitly report Laplacian definition (combinatorial vs normalized vs distance-weighted) and connectome construction choices, as these define the coordinate system. [18]
- For traveling waves, quantify detection sensitivity to source reconstruction, bandpass choice, and phase-estimation method; traveling-wave direction differs across modalities/resolutions. [93]

Core metrics to compute (aligned with the theory): - Modal coordinates: $a_k(t) = U^\top x(t)$ for SC-Laplacian or geometric eigenmodes; summarize by (i) power spectrum over k, (ii) entropy of modal energy distribution, (iii) covariance and temporal autocorrelation of $a_k(t)$. [26]
- Metastable substates: LEiDA cluster occupancies, dwell times, transition matrices; compare pre/post treatment and responders/non-responders. [94]
- Energy landscape: fit pairwise maximum entropy for a small set of networks/ROIs (scalable with care); compute minima count, basin depth proxies, and transition graph measures. [95]
- Traveling waves: proportion time in wave-like patterns, mean directionality, speed, and linkage to FC gradients or instrength gradients. [96]

Statistical tests (practical and defensible): - Pre/post within-subject: paired tests with permutation/bootstrapping on derived metrics; control multiple comparisons across modes k and networks (FDR). [97]
- Between-group: mixed-effects models with covariates (motion, site, medication), plus nonparametric validation where feasible. [98]
- Spatial autocorrelation-aware tests ("spin tests") when correlating cortical maps (e.g., mode weights or inferred targets with receptor density/gradients). [99]

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flowchart TD
  A[Acquire multimodal data: dMRI + rs-fMRI +/- MEG/EEG] --> B[Preprocess and QC]
  B --> C[Build structural connectome; choose Laplacian L]
  C --> D[Compute eigenmodes ψ_k, eigenvalues λ_k]
  B --> E[Compute functional signals x(t)]
  D --> F[Project to modal coordinates a_k(t)=<x,ψ_k>]
  E --> G[Compute FC and dFC: sFC, LEiDA, HMM, etc.]
  F --> H[Modal metrics: energy spectrum, entropy, variability]
  G --> I[State metrics: occupancy, dwell times, transitions]
  H --> J[Landscape inference in modal space: quasi-potential / stability proxies]
  I --> J
  J --> K[Compare groups & treatments; responders vs non-responders]
  K --> L[Fit mechanistic whole-brain model; perturb parameters/nodes in silico]
  L --> M[Generate testable predictions: stimulation targets, wave regime shifts]
```

This pipeline is directly motivated by eigenmode-basis methods (connectome harmonics; graph neural fields; geometric eigenmodes), metastable substate modeling, and clinical translation via perturbation analyses. [100]

Technical appendices with key equations

Static FC and Fisher transform

$$F_{ij} = \mathrm{corr}(x_i, x_j), \qquad z_{ij} = \mathrm{arctanh}(F_{ij})$$

Static FC is the baseline comparator for most depression meta-analyses and treatment studies. [101]

Instantaneous phase coherence (LEiDA-style)
Using analytic signals $x_n(t) = A_n(t)\cos(\phi_n(t))$, define

$$\mathrm{dPC}_{nm}(t) = \cos(\phi_n(t) - \phi_m(t)),$$

then take the leading eigenvector $v_1(t)$ of $\mathrm{dPC}(t)$ as the dominant phase-locking pattern. [102]

Graph Laplacian and graph Fourier transform
For an undirected graph with adjacency A and degree D,

$$L = D - A, \qquad LU = U\Lambda, \qquad \hat{x}(t) = U^{\top}x(t), \qquad x(t) = U\hat{x}(t).$$

This is the backbone for connectome harmonics and graph neural fields. [20]

Connectome harmonic decomposition (modal weights as state coordinates)

$$x(t) = \sum_{k=1}^{N} a_k(t)\psi_k, \quad a_k(t) = \langle x(t), \psi_k \rangle,$$

with "power/energy over modes" used as a state descriptor in psychedelic work. [35]

Geometric eigenmodes (Laplace–Beltrami) and wave model link

$$\Delta\psi = -\lambda\psi,$$

and dynamics modeled via damped wave equations on the cortical surface, with activity reconstructed by sums of eigenmodes. [41]

Hopf (Landau–Stuart) local dynamics in whole-brain models
A canonical local model used for fMRI whole-brain dynamics is the supercritical Hopf normal form, where a bifurcation parameter controls whether local dynamics decay to a fixed point or express oscillations; coupling through SC yields whole-brain regimes and transition behavior. [103]

Maximum entropy (pairwise Ising) energy landscape
For binary activity patterns $s \in \{-1, +1\}^N$:

$$E(s) = -\sum_i h_i s_i - \sum_{i<j} J_{ij} s_i s_j, \qquad p(s) \propto e^{-E(s)}.$$

Local minima of $E(s)$ define attractor-like basins; transitions encode barrier structure. [5]

Landscape curvature / precision link (psychedelic theory)
A central technical claim in REBUS is that decreasing the precision (confidence) of high-level beliefs corresponds to decreasing the curvature (flattening) of the relevant free-energy landscape, enabling escape from basins of attraction. [24]

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